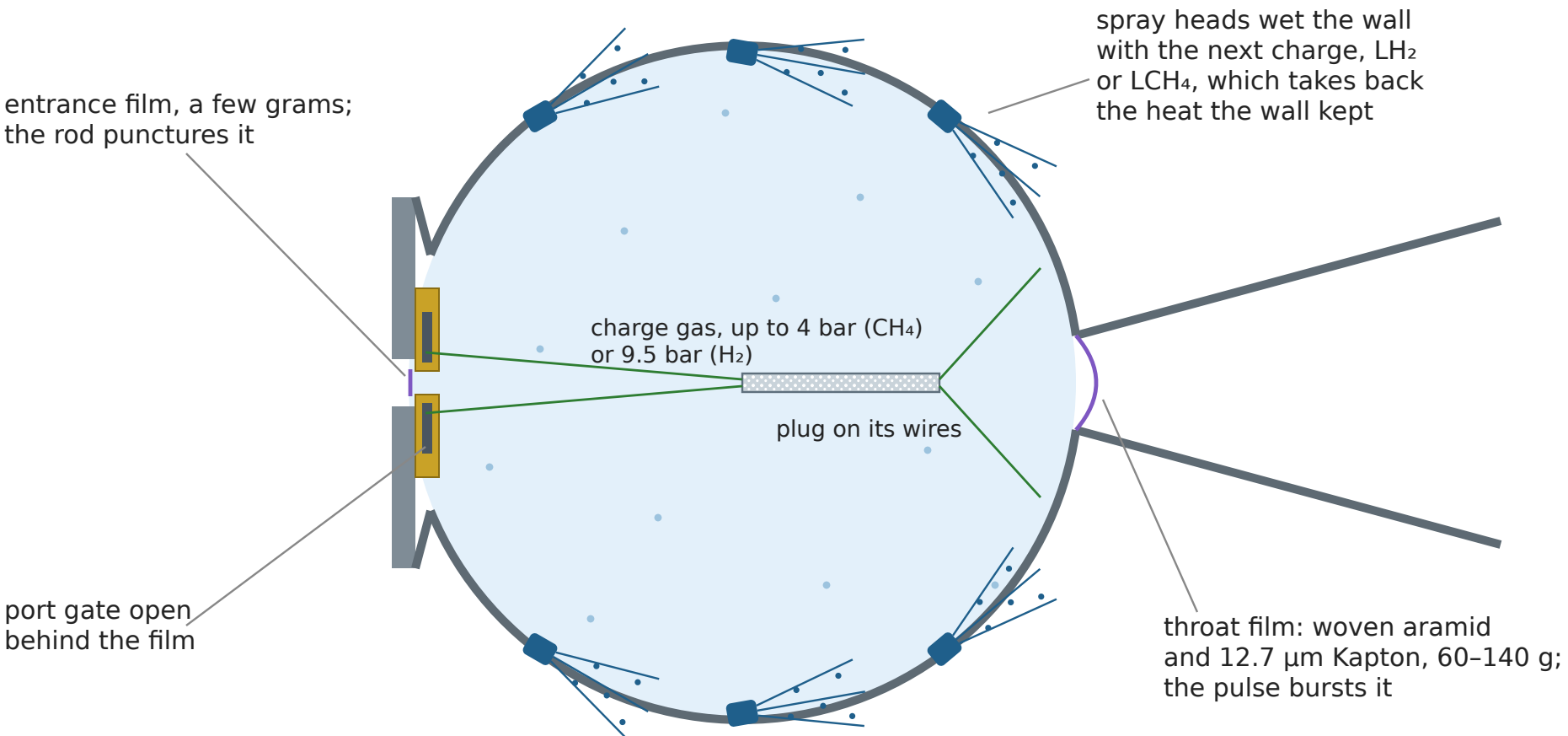
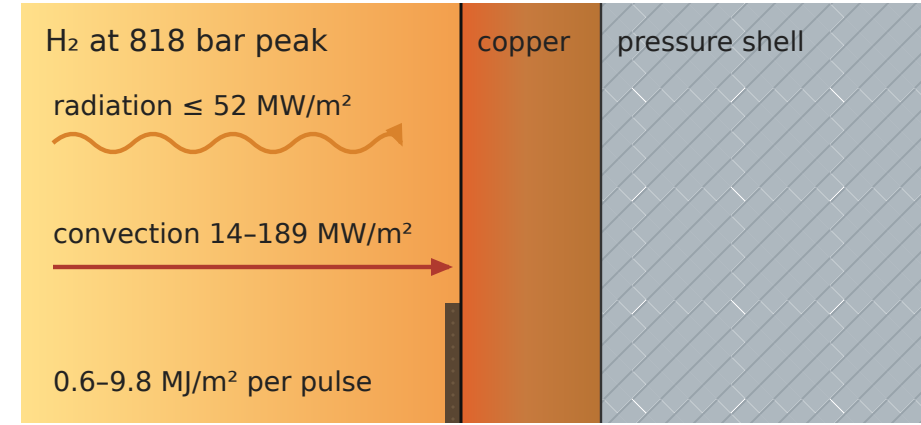


(a) Between pulses: spray the wall, hold the gas in



No coolant passages. The charge boils as it takes up the heat, so the chamber fills with gas and the films hold it until the next rod.

(b) Hydrogen, 5500 K: bare copper



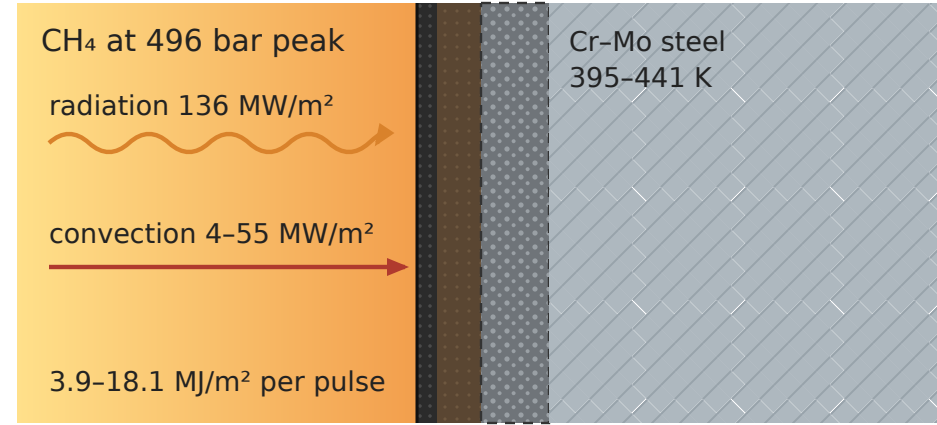
Copper conducts the heat inward faster than it builds, so its surface peaks at 370-1110 K, below its 1356 K melting point.

An LH₂ spray takes 21-350 MJ per pulse; warming the charge to room temperature absorbs about 230 MJ.

Pitch only on a hot band (dark strip), under 0.4 kg a pulse; hydrogen strips carbon.

Layer thicknesses are not to scale.

(c) Methane, 7000 K: pitch on steel



0.2 mm of pitch chars and ablates at ~3900 K, losing 30-120 μm (1.4-5.6 kg) a pulse; a 0.1 mm recoat replaces it.

An LCH₄ spray takes the 28-50 MJ the wall keeps, 4-7 MW/m² between pulses.

Dashed: optional porous graphite base, 1-2 mm. Graphite alone keeps 110-243 MJ, more than the charge can carry away.